SCIENCE EXPLORER / GRADE 8

Movement Into and Out of Cells

Watch. Predict. Explain.

Original revision guide with 100 practice questions and answer explanations.

Original teaching resource informed by KICD's revised 2024 Grade 8 curriculum. Five introductory missions for strand 2.0, sub-strand 2.2 Movement of Materials in and out of the Cell. The curriculum allocates 16 lessons to this sub-strand; this is not a full replacement. Independent prototype, not KICD-endorsed. Teacher review required. The source is linked, never bundled.

KICD Integrated Science Grade 8, revised 2024. strand 2.0, sub-strand 2.2 Movement of Materials in and out of the Cell. Source viewer pages 23-24 of 37. Newly written explanations, questions and visuals, not an official assessment or a copy of the source.

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View the original curriculum on Google Drive (external)

Virtual models only. Do not taste samples, smell unknown substances, handle biological specimens or heat sealed containers. Real tubing, dye and plant-material investigations require a teacher-approved method.

Use the HTML animation first, try its prediction, then read the optional explanation. All 100 practice questions are below. Short answers are self-assessed; equivalent scientifically correct wording is acceptable. No personal health information should be entered.

Your learning route

1. Diffusion: spread the idea: Explain net diffusion using random particle motion.

2. A selective boundary: Explain that membranes allow some substances through more readily than others.

3. Osmosis: water on the move: Predict net water movement in a simple two-solution model.

4. Cells gain or lose water: Connect water movement with simple plant-cell changes.

5. Transport keeps life going: Relate diffusion and osmosis to gas exchange and water uptake.

1. Diffusion: spread the idea

Explain net diffusion using random particle motion.

Diffusion is the net spreading of particles from a region of higher concentration to a region of lower concentration because of random motion. Individual particles can move in different directions; the overall pattern is the net movement.

A drop of suitable dye can spread through still water without stirring. Mixing caused by stirring or currents can speed distribution, but diffusion does not require either. Gases can also diffuse.

At equilibrium, there is no overall concentration-driven net movement, but particles do not stop moving. The animation shows a simplified, time-compressed distribution rather than tracking real collisions.

For otherwise similar conditions, higher temperature often speeds diffusion because particles move more rapidly on average. A greater concentration difference can increase the net rate. Use supplied data or a teacher-approved method rather than unknown dyes or vapours.

Visual mission

A crowd spreads out

Random motion can produce a net concentration change.

Time-compressed model, not real diffusion timings or collision paths.

Think it through

Explain why 'particles stop moving when evenly spread' is wrong.

2. A selective boundary

Explain that membranes allow some substances through more readily than others.

The cell membrane regulates movement into and out of a cell. It is selectively permeable: some substances can cross more readily than others, depending on their properties and the membrane.

A model with small blue water particles and larger amber solute markers can show a boundary that allows water but not that solute. The real reason for membrane selectivity is more complex than a literal kitchen sieve.

A partially permeable membrane is important in the introductory osmosis model. It separates solutions while allowing water to move. The selected solute stays on its side in that model.

Do not assume every substance enters by simple diffusion. Real cells also use other transport mechanisms. Teacher-approved tubing demonstrations are models, not exact copies of a living membrane.

Visual mission

A boundary with a difference

This model lets water through but blocks the selected solute.

A real membrane is more complex than a visible sieve. Blue means water; amber means the selected blocked solute.

Think it through

Explain why the virtual water can cross while the selected solute markers remain on their side.

3. Osmosis: water on the move

Predict net water movement in a simple two-solution model.

Osmosis is the net movement of water across a partially permeable membrane. In the introductory model, water moves overall from the more dilute solution towards the more concentrated solution when the selected solute cannot cross and other relevant conditions are comparable.

More dilute means a lower concentration of the selected solute, not a smaller volume. Water can move in both directions; the greater net transfer is towards the side with lower water potential under the stated conditions.

The animation conserves the total number of model water particles while redistributing them. Amber solute markers do not cross. It exaggerates visible movement and does not calculate real pressures or timings.

When the relevant conditions balance, net water transfer can be zero while water molecules continue moving. Pressure differences can affect real osmosis, so 'water always moves towards more solute in every possible situation' is too broad.

Visual mission

Water goes both ways. Count the net.

In this simple model, net water moves towards the more concentrated solution.

The selected solute cannot cross; pressures and other conditions are simplified. Particle counts, not real timing or volume, are shown.

Think it through

Predict the net direction when two solutions are separated by a water-permeable, solute-impermeable membrane.

4. Cells gain or lose water

Connect water movement with simple plant-cell changes.

A plant cell placed in a suitable more dilute surrounding solution can gain water by osmosis. Its vacuole can enlarge, and pressure against the supporting wall helps make the cell turgid.

In a suitable more concentrated surrounding solution, a plant cell can lose water. Its contents can shrink away from the wall in a sufficiently strong case, a process called plasmolysis.

The cell wall and membrane are different: the rigid wall helps support the cell, while the membrane remains the selective boundary. The virtual cell's size changes are exaggerated for visibility.

These predictions assume the relevant solute is not freely crossing and conditions suit the simple model. They are not instructions to treat people, animals or plants with arbitrary salt solutions.

Visual mission

A plant cell gains or loses water

Water balance affects the cell contents inside the supporting wall.

Exaggerated plant-cell model. The selected solute does not freely cross; this is not a treatment for living things.

Think it through

Use the surrounding-solution selector and explain what changes in the model cell and what remains a wall.

5. Transport keeps life going

Relate diffusion and osmosis to gas exchange and water uptake.

In the lungs, oxygen can diffuse from air spaces into blood where its concentration is lower, while carbon dioxide can diffuse in the opposite overall direction under normal conditions.

A thin exchange surface and a large surface area can support faster exchange. Breathing and blood flow help maintain differences, but bulk ventilation and diffusion are not the same process.

Water can enter root cells by osmosis under suitable conditions. Mineral uptake is not itself osmosis: osmosis refers to water, and cells can use other mechanisms for solutes.

Absorption in the intestine includes several mechanisms. Some substances can diffuse, but not every nutrient enters only by simple diffusion. Use the model to explain selected processes rather than all of biology.

Visual mission

Two gases. Different directions.

Name the substance before explaining its net direction.

A simplified normal-condition gas-exchange model, not a diagnostic tool or every intestinal transport mechanism.

Think it through

Explain why 'all nutrients enter by osmosis' confuses the moving substance with other transport processes.

Revision cards

Diffusion

Net spreading from higher to lower concentration due to random motion.

Equilibrium

No overall net transfer; particles can still move.

Concentration

Amount of a specified substance per volume.

Selective membrane

Some substances cross more readily than others.

Osmosis

Net water movement across a partially permeable membrane.

Dilute solution

Lower concentration of the selected solute.

Turgid cell

Water-filled plant cell pressing against its wall.

Plasmolysis

Contents pull away from the wall after sufficient water loss.

Gas exchange

Oxygen and carbon dioxide can diffuse in different net directions.

Model limit

Not all nutrient or mineral uptake is simple diffusion or osmosis.

100-question practice bank

40 multiple-choice and 60 short-answer questions. This is a practice bank, not a standardised exam.

Diffusion: spread the idea

1. What causes diffusion?

A. A need to stir every time

B. Particles choosing a destination

C. Creation of new matter

D. Random particle motion

2. Net diffusion tends to go from where to where?

A. Only downwards

B. Only towards light

C. Higher to lower concentration

D. Lower to higher in every case

3. Does diffusion require stirring?

A. Only for gases

B. No

C. Always

D. Only in a labelled cup

4. At equilibrium, particles do what?

A. Keep moving with no overall net concentration-driven transfer

B. Stop completely

C. Disappear

D. Become larger

5. Which can undergo diffusion?

A. Only solid walls

B. Only hot objects

C. Only living cells

D. Particles in liquids and gases

6. Higher temperature often affects diffusion how, with other factors similar?

A. It removes particles

B. It turns all water into salt

C. It speeds it up

D. It always stops it

7. What does net movement mean?

A. Only visible large-object movement

B. The overall movement after opposite movements are considered

C. Every particle moving the same way

D. No movement at all

8. Which is a safe approach to an unknown vapour?

A. Do not smell it; tell the teacher

B. Sniff it closely

C. Taste nearby liquid

D. Heat it at home

9. Define diffusion.

10. Give a diffusion example in a liquid.

11. Give a diffusion example in a gas.

12. Why is stirring not part of the definition?

13. What happens to particle motion at equilibrium?

14. Explain higher concentration.

15. Why can a stronger concentration difference increase net diffusion?

16. How does temperature influence average particle motion?

17. Why is a time-compressed animation not experimental timing data?

18. Does every particle travel straight from high to low concentration?

19. What should be controlled in a temperature comparison?

20. What is wrong with saying equilibrium means no particles exist?

A selective boundary

21. What does selectively permeable mean?

A. Everything passes equally

B. Nothing can ever cross

C. The membrane is absent

D. Some substances cross more readily than others

22. Which boundary is central to the introductory osmosis model?

A. A metal ruler

B. A flame

C. A partially permeable membrane

D. An open empty space only

23. What do blue particles represent in this model?

A. Microscope lenses

B. Water

C. All possible solutes

D. Atoms of copper

24. What do amber markers represent here?

A. The selected non-crossing solute

B. Every molecule in a real cell

C. Air bubbles only

D. Water itself

25. Is a real membrane exactly a kitchen sieve?

A. Always

B. Only when magnified

C. Only in animals

D. No

26. Does every substance enter cells only by simple diffusion?

A. Only in plants

B. Only in animals

C. No

D. Always

27. Which structure controls cell entry and exit?

A. A classroom wall

B. Cell membrane

C. Slide label

D. Microscope eyepiece

28. What should guide a real tubing demonstration?

A. A teacher-approved method

B. An improvised unknown chemical mixture

C. Tasting the contents

D. Heating a sealed bag

29. Describe a selectively permeable membrane.

30. Why is a colour key important in the membrane model?

31. What stays on its side in this osmosis model?

32. Which substance can cross in the model?

33. Name one limitation of representing a membrane as holes.

34. Why is a cell wall not a substitute for explaining the membrane?

35. Does selective movement mean particles make decisions?

36. Why should a tubing model not be called a complete living cell?

37. What must be known before predicting which particles cross?

38. Why is an arrow only part of the explanation?

39. What should a learner do with unknown model liquids?

40. Why mention other cell-transport processes?

Osmosis: water on the move

41. Which substance moves in osmosis?

A. Only the blocked solute

B. The membrane itself

C. A microscope lens

D. Water

42. What boundary is required in this model?

A. A flame

B. A dry metal sheet

C. A partially permeable membrane

D. No boundary at all

43. Under the stated simple conditions, net water moves towards which solution?

A. Always the warmer label

B. The more concentrated solution

C. Always the smaller volume

D. Always the left side

44. More dilute means what?

A. Lower solute concentration

B. Always less total liquid

C. Higher solute concentration

D. No water

45. Can water move both ways across the membrane?

A. Never

B. Only before the page loads

C. Only if stirred

D. Yes

46. What happens to the selected solute in this animation?

A. It crosses freely

B. It disappears

C. It remains on its side

D. It all becomes water

47. Does the model create new water particles?

A. Only at equilibrium

B. No

C. Always

D. Only on the right

48. Which claim is too broad?

A. Water always moves towards more solute regardless of all other conditions

B. The membrane matters

C. Pressure can affect real osmosis

D. Water can move both ways

49. Define osmosis for this lesson.

50. State the predicted net direction in the simple dilute-to-concentrated setup.

51. Why is 'towards the side with more liquid' not a definition?

52. What does a large amber marker stand for?

53. What should happen to the model's total water count?

54. What does zero net water transfer mean?

55. Why are arrows drawn in both directions in some osmosis models?

56. Why must the membrane assumptions be stated?

57. Name another factor that can affect real osmosis.

58. Does adding more dilute solution always mean adding more solute concentration?

59. How could you check the animation's water conservation?

60. Why is osmosis not a particle's purposeful choice?

Cells gain or lose water

61. A plant cell in a suitable more dilute solution tends to do what?

A. Always lose all water

B. Lose its cell wall immediately

C. Become a chemical element

D. Gain water

62. A plant cell in a suitable more concentrated solution tends to do what?

A. Become metal

B. Stop having particles

C. Lose water

D. Create water

63. What contributes to a turgid plant cell's support?

A. A lack of membrane

B. Water-related pressure against its wall

C. A microscope battery

D. A flame

64. Which structure stays as the supporting outer wall in the model?

A. Cell wall

B. Cell sap only

C. Nucleus only

D. Eyepiece

65. What is plasmolysis?

A. Water creating a new nucleus

B. A type of fire

C. Metal rusting

D. Cell contents shrinking away from the wall after water loss

66. Are the model's size changes exact biological measurements?

A. Only when coloured

B. Only on a phone

C. No

D. Always

67. Which assumption matters for this simple prediction?

A. Cells contain no water

B. The relevant solute does not freely cross

C. All solutes behave identically

D. Pressure never matters anywhere

68. Is the animation a reason to apply arbitrary salt mixtures to living things?

A. No

B. Yes, always

C. Only if the result looks dramatic

D. Only at home

69. Explain turgid in the plant-cell context.

70. Describe a model cell gaining water.

71. Describe a model cell losing water.

72. Why is the wall not the same as the selective membrane?

73. What does the surrounding-solution selector change?

74. Why should the word tends appear in simple predictions?

75. What remains after the illustrated contents pull inward?

76. Why can water balance affect plant firmness?

77. Does a shrunken diagram prove a real sample was observed?

78. How can learners investigate safely before practical work?

79. What should be compared when predicting water entry?

80. Why is this not a medical treatment lesson?

Transport keeps life going

81. In normal lung gas exchange, oxygen can diffuse overall from where?

A. Blood into bone only

B. A microscope lens into air

C. Nowhere

D. Air spaces into blood

82. Carbon dioxide in normal lung exchange can diffuse overall towards where?

A. A new metal

B. The classroom floor

C. The lung air spaces

D. A solid cell wall only

83. Which process specifically names water movement across a partially permeable membrane?

A. Magnification

B. Osmosis

C. All nutrient absorption

D. Burning

84. Does mineral uptake itself mean osmosis?

A. No

B. Always

C. Only for blue minerals

D. Only in animals

85. What can a thinner exchange surface do, all else similar?

A. Create new oxygen

B. Remove every membrane

C. Stop particle movement

D. Shorten the diffusion distance

86. What can a larger exchange area support?

A. No concentration difference

B. No need for water

C. More exchange under suitable conditions

D. Guaranteed infinite exchange

87. Is breathing identical to diffusion?

A. Only in plants

B. No

C. Always

D. Only during daytime

88. Which statement about intestinal absorption is careful?

A. It involves more than one transport mechanism

B. Every nutrient enters by osmosis

C. No cells are involved

D. All nutrients are water

89. Give one role of diffusion in the lungs.

90. Give one role of osmosis in a plant.

91. Why is mineral uptake not called osmosis?

92. Explain a thin exchange surface's advantage.

93. Explain a large exchange surface's advantage.

94. How can blood flow help gas exchange?

95. How can ventilation help gas exchange?

96. Why should a nutrient-absorption explanation avoid the word all with simple diffusion?

97. Name the gas exchanged out of blood towards lung air in the model.

98. What is a concentration gradient?

99. Why should labelled arrows name the substance?

100. How can a learner explain transport without unsafe experiments?

Answer key & explanations

1. D. Random particle motion

Net spreading emerges from random movement.

2. C. Higher to lower concentration

This describes the overall concentration-driven trend.

3. B. No

Particles move randomly without stirring.

4. A. Keep moving with no overall net concentration-driven transfer

No net movement does not mean no motion.

5. D. Particles in liquids and gases

Diffusion is not restricted to cells.

6. C. It speeds it up

Average particle motion increases.

7. B. The overall movement after opposite movements are considered

Individual motion and overall transfer differ.

8. A. Do not smell it; tell the teacher

Diffusion examples do not justify exposure.

9. Net spreading from higher to lower concentration due to random particle motion.

The definition describes an overall effect.

10. A suitable dye spreading through still water.

Use only an approved demonstration or model.

11. A vapour spreading through air.

Do not deliberately smell unknown substances.

12. Particles move randomly even without bulk mixing.

Stirring and diffusion are different processes.

13. It continues.

Opposite movements balance on average.

14. More of the relevant particles in a given volume.

Specify which particles are being compared.

15. It creates a greater imbalance in particle movements across the region.

Other conditions must be comparable.

16. Higher temperature generally increases average kinetic energy.

The particles themselves do not become larger.

17. Its speed is chosen for teaching rather than measured.

It cannot predict a real waiting time.

18. No.

Random individual paths produce the net trend.

19. Other factors such as substance, concentration difference and setup.

A fair comparison isolates the tested factor.

20. Particles remain and continue moving.

Equilibrium concerns net transfer, not disappearance.

21. D. Some substances cross more readily than others

Membrane properties affect movement.

22. C. A partially permeable membrane

Water can cross while the selected solute does not.

23. B. Water

Always read the diagram key.

24. A. The selected non-crossing solute

The model specifies which substance cannot cross.

25. D. No

The sieve comparison has limits.

26. C. No

Cells have other transport mechanisms too.

27. B. Cell membrane

It regulates exchanges with the surroundings.

28. A. A teacher-approved method

Models do not remove practical hazards.

29. A boundary across which some substances pass more readily than others.

It is not simply an unrestricted opening.

30. It identifies water and the selected solute.

Colour does not measure real particle size or appearance.

31. The selected solute.

The model assumes it cannot cross the membrane.

32. Water.

The relevant boundary is partially permeable.

33. Real selectivity depends on more than simple visible hole size.

The diagram is not a literal membrane photograph.

34. They are different structures; the membrane regulates exchanges.

Plant cells have both.

35. No.

Physical properties and interactions explain movement.

36. It lacks the full structures and processes of a cell.

It demonstrates selected transport ideas.

37. The substance and the boundary's permeability.

Do not assume every membrane behaves identically.

38. It shows a direction but not every individual particle path or mechanism.

Interpret it with the key and assumptions.

39. Avoid tasting or handling them and ask the teacher.

Use the virtual model instead.

40. To avoid falsely claiming all absorption is simple diffusion.

An introductory model has a defined scope.

41. D. Water

This lesson's definition concerns net water movement.

42. C. A partially permeable membrane

The boundary allows water but not the selected solute.

43. B. The more concentrated solution

Use solute concentration and model assumptions.

44. A. Lower solute concentration

Concentration is not the same as volume.

45. D. Yes

Net movement is the difference between opposite movements.

46. C. It remains on its side

The model assumes the membrane blocks that solute.

47. B. No

It redistributes a conserved model total.

48. A. Water always moves towards more solute regardless of all other conditions

Predictions need appropriate assumptions.

49. Net water movement across a partially permeable membrane.

Direction depends on water potential under the relevant conditions.

50. From the more dilute towards the more concentrated solution.

The selected solute cannot cross and other conditions are comparable.

51. Volume alone does not determine the concentration difference.

Compare the amount of solute per volume.

52. A model solute particle that cannot cross this membrane.

It is a teaching key, not a literal colour or size.

53. It should stay constant.

Net redistribution is not water creation.

54. Opposite transfers balance overall.

Molecules can still move.

55. Water moves both ways even when one direction has the larger net transfer.

A single net arrow simplifies that movement.

56. Different permeability would change the prediction.

If the solute also crosses, the simple model may not apply.

57. Pressure differences.

The concentration-only picture is not universal.

58. No.

Dilute refers to lower concentration, not necessarily lower total amount.

59. Count water markers on both sides before and after.

The total should remain the same.

60. Movement follows physical interactions and differences in water potential.

Molecules do not decide where to go.

61. D. Gain water

Net water can enter by osmosis.

62. C. Lose water

Net water can leave under the stated assumptions.

63. B. Water-related pressure against its wall

The wall resists expansion.

64. A. Cell wall

The membrane and cell contents can move relative to it.

65. D. Cell contents shrinking away from the wall after water loss

It can occur in a sufficiently concentrated surrounding solution.

66. C. No

They are exaggerated illustrations.

67. B. The relevant solute does not freely cross

Permeability affects the result.

68. A. No

Use approved teaching models, not unsafe experiments.

69. A water-filled cell whose contents press against the supporting wall.

Turgor contributes to plant support.

70. Its vacuole and contents expand within the wall.

The illustration is exaggerated.

71. Its contents shrink, potentially pulling away from the wall.

A strong enough case can show plasmolysis.

72. They are different structures with different roles.

The membrane regulates movement.

73. The relative solute concentration outside the model cell.

The model specifies the permeability assumptions.

74. Actual movement depends on conditions and permeability.

Avoid unlimited claims.

75. The model cell wall remains in place.

Its presence distinguishes the structures.

76. Turgor depends on water-related pressure in cells.

Plants also have other supporting structures.

77. No.

It is a generated teaching model.

78. Use the virtual selector and discuss predictions with the teacher.

Real specimens need approved methods.

79. Inside and outside solution conditions across the membrane.

A single colour without a key is insufficient.

80. It explains a cell model, not how to treat a person.

Health decisions need appropriate professional guidance.

81. D. Air spaces into blood

The direction follows the relevant concentration difference.

82. C. The lung air spaces

It can move from blood towards air before exhalation.

83. B. Osmosis

Keep the moving substance clear.

84. A. No

Osmosis concerns water, not all solutes.

85. D. Shorten the diffusion distance

Distance can affect exchange rate.

86. C. More exchange under suitable conditions

Area is one factor, not the only one.

87. B. No

Breathing moves bulk air; diffusion is particle-level net movement.

88. A. It involves more than one transport mechanism

Do not overextend the simple model.

89. Oxygen moving from air spaces into blood under normal conditions.

Carbon dioxide exchange is another example.

90. Water entering root cells under suitable conditions.

The membrane and water-potential difference matter.

91. Osmosis refers specifically to water movement.

Solutes may use other mechanisms.

92. It reduces the distance particles must diffuse.

Other factors still affect rate.

93. It provides more area over which exchange can occur.

This does not make exchange unlimited.

94. It helps maintain relevant concentration differences.

Bulk flow and diffusion work together but differ.

95. It refreshes air in the lungs and helps maintain differences.

It is not itself the definition of diffusion.

96. Several transport mechanisms can be involved.

The simple model covers selected cases.

97. Carbon dioxide.

The model assumes normal gas-exchange conditions.

98. A difference in concentration across space.

It helps describe a direction for net diffusion.

99. Different substances can have different net directions.

Oxygen and carbon dioxide need not move the same way.

100. Use diagrams, simulations and teacher-supplied data.

Avoid unknown substances and biological samples.

Explore responsibly

The HTML includes YouTube topic-search links, not teacher-vetted individual videos. External content may include advertising, inaccurate claims or age-inappropriate recommendations. A teacher should preview results. The original built-in animations need no internet.

Share the complete site folder, or host it with index.html at the root. Progress is stored locally and separately for each topic; there are no accounts or shared class records.

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